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Assignement :

Variant 1 :

We have : $x(t) = \lambda \int_{0,3}^{1,2} t^2 s^2 x(s)\,ds + \frac{\sin t}{t}$

- $\lambda = 0,6$; $a = 0,3$; $b = 1,2$;
- $K(t,s) = t^2 s^2$;
- $y(t) = \frac{\sin t}{t}$

Part 1 : the Contraction :

$$(Ax)(t) = \lambda \int_a^b K(t,s)\,x(s)\,ds + y(t)$$

- the metric space is : $C[a,b]$

where : $\rho(x,z) = \max_{a \le t \le b} |x(t) - z(t)|$

for x, z : $|Ax(t) - Az(t)|$

equals : $\left| \lambda \int_a^b t^2 s^2 (x(s) - z(s))\,ds \right|$

then we use : $|x(s) - z(s)| \le \rho(x,z)$

we obtain : $|Ax - Az| \le |\lambda|\, t^2 \rho(x,z) \int_a^b s^2\,ds$

$\Rightarrow \int_{0,3}^{1,2} s^2\,ds = \frac{(1,2)^3 - (0,3)^3}{3} =$

$\Rightarrow (1,2)^3 = 1,728$; $(0,3)^3 = 0,027$

then : $\int s^2\,ds = \frac{1,701}{3} = 0,567$

So the maximum of t^2 on $[0,3 , 1,2]$ is : $1,41$

and : $\theta = |\lambda| \cdot 1,41 \cdot 0,567 =$

$= 0,6 \cdot 1,41 \cdot 0,567 = 0,4808$

$$\boxed{\theta \simeq 0,48 < 1}$$

The operator is a Contraction so the theorem applies.

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Part 2 : The Exact Solution :

We have : $K(t,s) = t^2 \cdot s^2$

we can do : $I = \int_{0,3}^{1,2} s^2 x(s)\,ds$

and : $x(t) = 0,6t^3 I + \frac{\sin t}{t}$

Let's integrate : $I = 0,6\,I \int_{0,3}^{1,2} s^4\,ds + \int_{0,3}^{1,2} s \sin s\,ds$

so : 1st $\int s^4\,ds = \frac{s^5}{5} =$

$\Rightarrow (1,2)^5 = 2,488 \; ; \; (0,3)^5 = 0,002$

then : $\int s^4\,ds = \frac{2485}{5} = 0,497$

$\Rightarrow 0,6 \cdot 0,497 = 0,298$

2nd $\int s \sin s\,ds = -s \cos s + \sin s$

Evaluation :

at 1,2 : $-1,2 \cos 1,2 + \sin 1,2 = 0,497$

at 0,3 : $-0,3 \cos 0,3 + \sin 0,3 = 0,008$

* Difference is : 0,488

then : $I = 0,298 + 0,4889$

so : $I = \frac{0,4889}{0,7016} = 0,696$

$\Rightarrow 0,6\,I = 0,418$

The final and exact Solution :

$$\boxed{x(t) = 0,4180\,t^2 + \frac{\sin t}{t}}$$

Part 3 : Calculation of L :

- The first : $x_1(t) = \frac{\sin t}{t}$
- The Second : $x_2(t) = 0,6t^2 \int_{0,3}^{1,2} s^2 \frac{\sin s}{s} ds \cdot \frac{\sin t}{t}$

and : $\int s \sin s \, ds = 0,4889$

so : $x_2 = 0,293 t^2 + \frac{\sin t}{t}$

$\Rightarrow p(x_1, x_2) = \max 0,293 t^2$

the maximum occurs at : $t = 1,2$

$\Rightarrow p(x_1, x_2) = 0,293 \cdot 1,44 = 0,422$

then : $\frac{\theta^{L-1}}{1-\theta} p(x_1, x_2) < 0,01 \quad ; \theta = 0,489 \;;$

and $1 - \theta = 1 - 0,489 = 0,510$

after that : $0,827 \theta^{L-1} < 0,01 \; =$

$\Rightarrow \theta^{L-1} < 0,01208 \; =$

$\Rightarrow L - 1 > \frac{\ln 0,01208}{\ln 0,489} = 6,27$

After that : $\boxed{L = 8}$

Part 4 : Python implementation : (attached you can find it).

```
8
9  import numpy as np
10
11 # Parameters
12 a = 0.3
13 b = 1.2
14 lam = 0.6
15 n = 9
16
17 N = 2 * n
18 h = (b - a) / N
19
20 t = np.linspace(a, b, N + 1)
21
22 # Right-hand side
23 y = np.sin(t) / t
24
25 # Initial approximation
26 x = y.copy()
27
28 # Simpson weights
29 w = np.ones(N + 1)
30 w[1:N:2] = 4
31 w[2:N-1:2] = 2
32
33 # Number of iterations
34 L = 8
```

```
x = y.copy()

# Simpson weights
w = np.ones(N + 1)
w[1:N:2] = 4
w[2:N-1:2] = 2

# Number of iterations
L = 8

for k in range(L):
    x_new = np.zeros_like(x)

    for i in range(N + 1):
        kernel = (t[i] ** 2) * (t ** 2)
        integral = (h / 3.0) * np.sum(w * kernel * x)
        x_new[i] = lam * integral + y[i]

    x = x_new

# Exact solution
exact = 0.41804 * t**2 + np.sin(t)/t

R = np.max(np.abs(exact - x))

print("L =", L)
print("R =", R)
```

```
45
46  # Exact solution
47  exact = 0.41804 * t**2 + np.sin(t)/t
48
49  R = np.max(np.abs(exact - x))
50
51  print("L =", L)
52  print("R =", R)
```

input

```
L = 8
R = 0.00077929287577638 28


...Program finished with exit code 0
Press ENTER to exit console.
```